

Data Storage

- Binary digit (bit) 
 - 8 bit $\longrightarrow$ 1 byte
 - memory
 - $(ABCD)_{16}$
-

logic gates

boolean operations

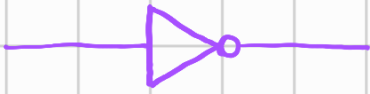
And

OR

XOR

Not

x	output
0	1
1	0



And

x	y	output
0	0	0
0	1	0
1	0	0
1	1	1



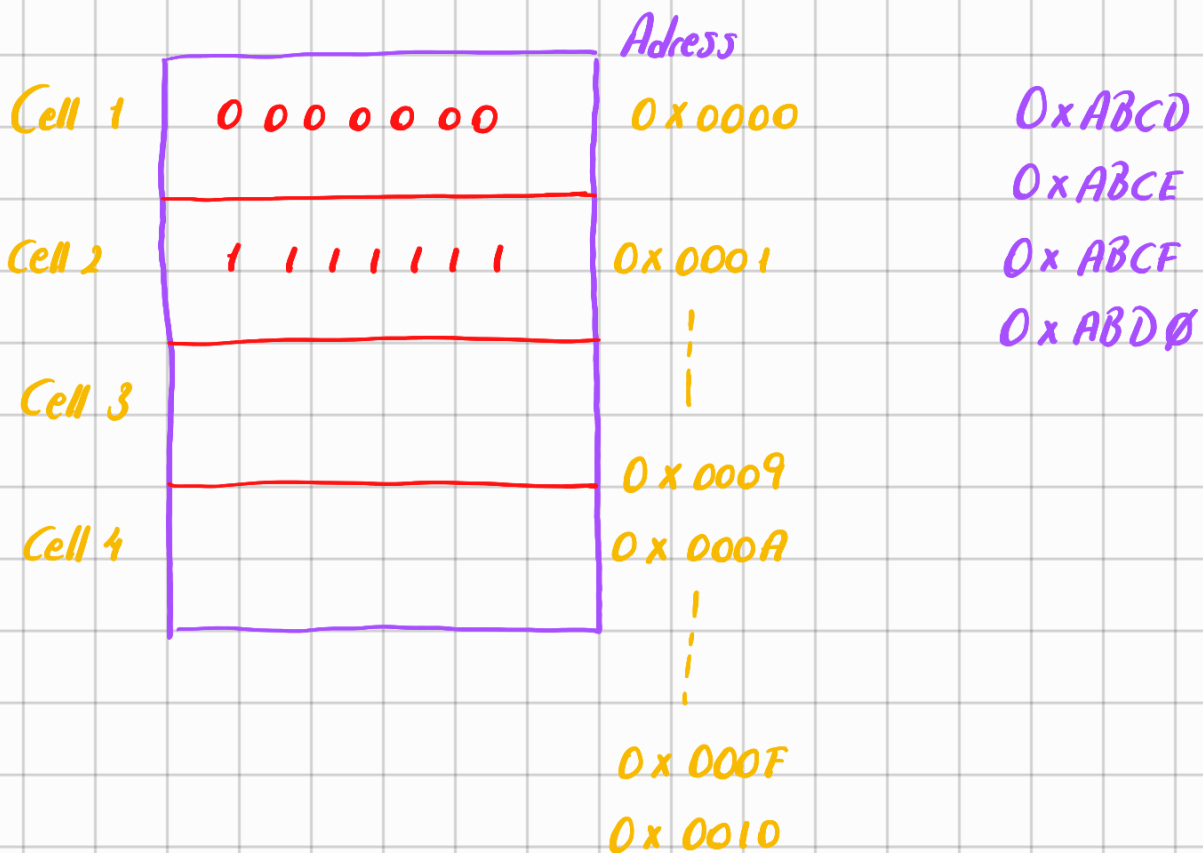
OR

x	y	output
0	0	0
0	1	1
1	0	1
1	1	1



XOR (exclusive OR)

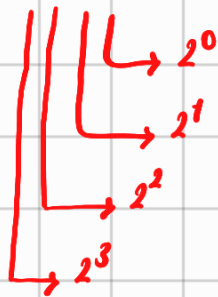
X	y	output
0	0	0
0	1	1
1	0	1
1	1	0



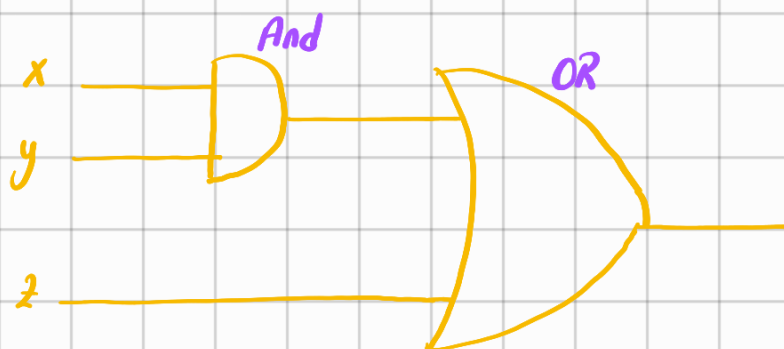
Main memory

01011100

Least significant bit
(low order end)

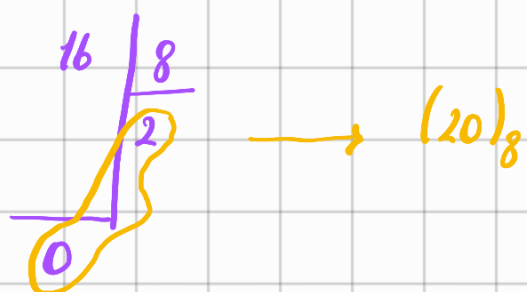
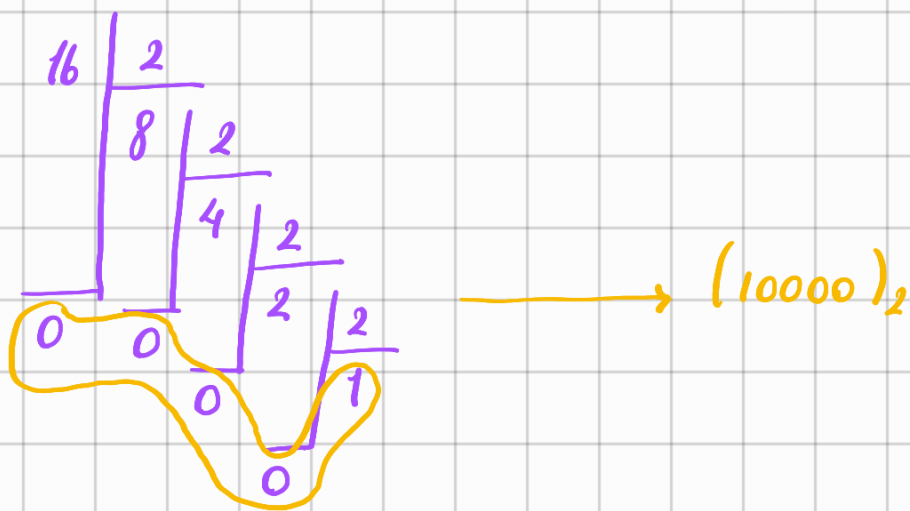


Most significant bit
(high order end)



Output

x	y	z	Output
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1



Unsigned number

$$(11100000)_2 \longrightarrow (224)_{10} \longrightarrow (340)_8 \longrightarrow (E0)_{16}$$

Diagram showing the conversion of the binary number $(11100000)_2$ to decimal $(224)_{10}$, octal $(340)_8$, and hexadecimal $(E0)_{16}$. The octal and hexadecimal results are annotated with their respective powers of 8 and 16:

- Octal: $3 \cdot 8^2 + 4 \cdot 8^1 + 0 \cdot 8^0$
- Hexadecimal: $E \cdot 16^1 + 0 \cdot 16^0$

$$(00010000)_2 \longrightarrow (16)_{10} \longrightarrow (20)_8 \longrightarrow (10)_{16}$$

$$(10000000)_2 \longrightarrow (128)_{10} \longrightarrow (200)_8 \longrightarrow (80)_{16}$$

Diagram showing the conversion of the binary number $(10000000)_2$ to decimal $(128)_{10}$, octal $(200)_8$, and hexadecimal $(80)_{16}$. The octal and hexadecimal results are annotated with their respective powers of 8 and 16:

- Octal: $2 \cdot 8^2 + 0 \cdot 8^1 + 0 \cdot 8^0$
- Hexadecimal: $8 \cdot 16^1 + 0 \cdot 16^0$

Represently information as bit patterns

Numbers

Charoctors (text)

'A' 'B' "ob"

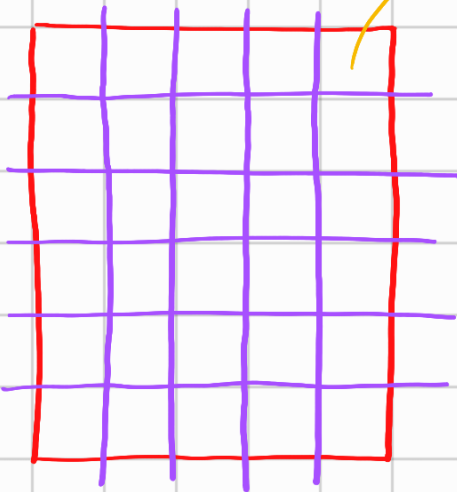
image

video

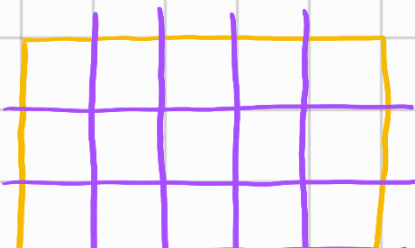
Sound

Image

→ pixel (picture element)



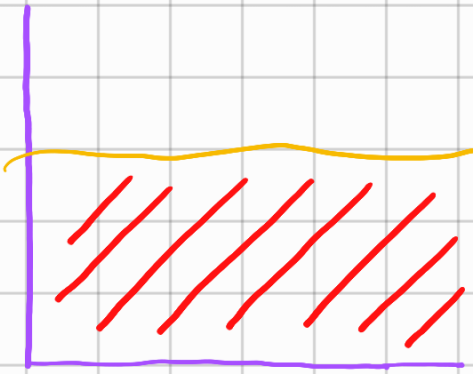
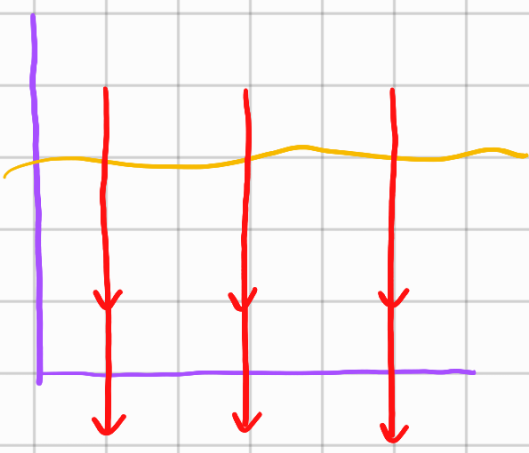
Video





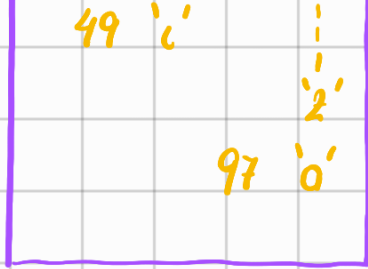
30 fps
240 fps

Analog vs digital



65 'o'
48 'o'

Ascii table
(American Standart Code information interchange)



ANSI

American National Standard Institute

1 byte $\rightarrow$ en fazla 255 sayısını olabilir 256 farklı sayı olabilir

UTF-8 Unicode transformation format 8-bit

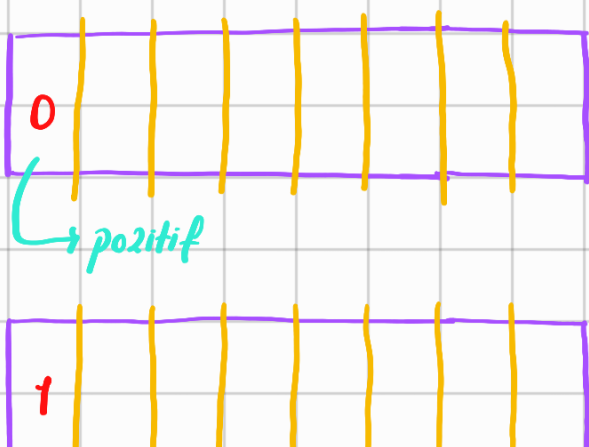
2 byte 65536 $\rightarrow 2^{16}$

Negative Numbers

Two's complement

Excess Notation

(-5) Two's complement notation



negatif

One's complement

0 0 0 0 1 1 1 1

1 1 1 1 0 0 0 0

0 1 0 1 1 0 0 1

1 0 1 0 0 1 1 0

Two's complement

- One's complement

- Odd 1

0 0 0 0 1 1 1 1

1 1 1 1 0 0 0 0

1 1 1 1 0 0 0 1

2's complement

0 1 0 1 1 0 0 1

1 0 1 0 0 1 1 0

1 0 1 0 0 1 1 1

0 0 1 1 1 1 0 0

1 1 0 0 0 0 1 1

1 1

1 1 0 0 0 1 0 0 2's Complement

Signed number system

$(01100000)_2 \rightarrow (96)_{10}$

$(10000000)_2 \rightarrow (-128)_{10}$

Negative

0 1 1 1 1 1 1 1

1 1

1 0 0 0 0 0 0 0

(11000001) → signed $(-63)_{10}$

→ unsigned $(193)_{10}$

→ Negative

0 0 1 1 1 1 1 0

1 1

0 0 1 1 1 1 1 1

Two's complement notation

011	3
010	2
001	1
000	0
111	-1
110	-2
101	-3
100	-4

0110



1010

0110

1001

+ 1

1010

11111111 → -8

00000000

0 0 0 0 0 0 0 0 $\rightarrow$ 0

Excess Notation

1 1 1	3	$\rightarrow$	7-4	$\rightarrow$	3
1 1 0	2	$\rightarrow$	6-4	$\rightarrow$	2
1 0 1	1	$\rightarrow$	5-4	$\rightarrow$	1
1 0 0	0	$\rightarrow$	4-4	$\rightarrow$	0
0 1 1	-1	$\rightarrow$	3-4		
0 1 0	-2	$\rightarrow$	2-4		
0 0 1	-3	$\rightarrow$	1-4		
0 0 0	-4	$\rightarrow$	0-4		

Excess -4 notation

(-4, 3)

Excess 8 notation

1 1 1 1	15-8	7
1 0 0 0	0	
0 0 0 0	0-8	-8

1 byte $\rightarrow$ unsigned (0, 255)

1 byte $\rightarrow$ signed (-128, 127)

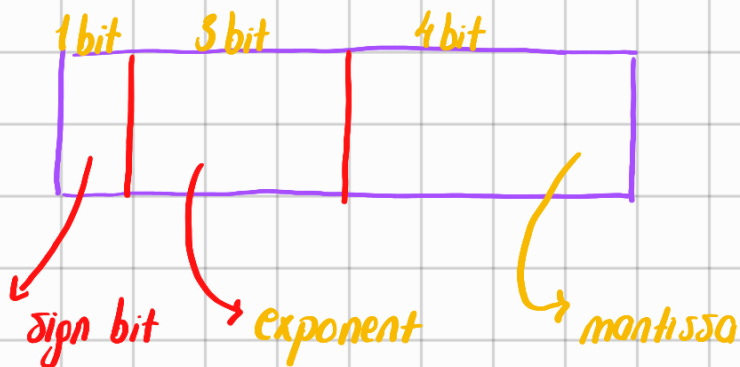
Storing fractions

$(5,25)_{10}$

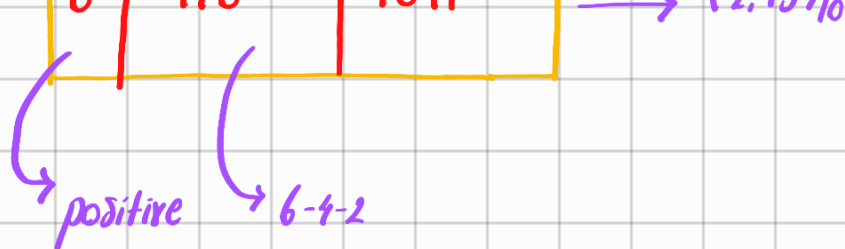
$(100,11)_2 \rightarrow 4,75 \rightarrow 4 \frac{3}{4} \rightarrow 4 + \frac{1}{2} + \frac{1}{4}$

$(10,01)_2 \rightarrow 2,25 \rightarrow 2 \frac{1}{4}$

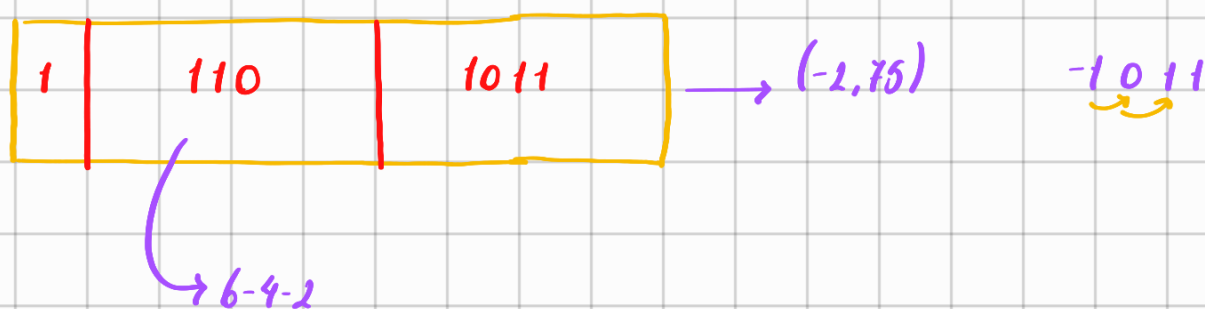
Floating point notation



0 | 110 | 1011 $\rightarrow (2,75)_{10}$



-1011 Notlayı 2 soro
ilerletiriz $10.11 \rightarrow 2 \frac{3}{4}$



$(010111100)_2$

$\hookrightarrow$ positive

011

$3-4 \rightarrow -1$

$.1100$

$.01100 \rightarrow \frac{1}{4} + \frac{1}{8} \rightarrow \frac{3}{8}$

$1 \frac{1}{2} \rightarrow (1.5)_{10} \rightarrow (01011100)_2 \rightarrow (1.5)$

1.1000

.1100

1.1 $\rightarrow (1.5)_{10}$

3/8

.011 $\rightarrow$ 1'i ilk gördüğün anda başla 0011
yapma 0'lar boşuna
yer kaplar

0	011	1100
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$\rightarrow$ 1'i sola kaymak
için 011 yazılır

-32 $\rightarrow$ 00100000

11011111

+ 1

-32 $\leftarrow$ 11100000

$$\begin{array}{r}
 -1 \rightarrow 00000001 \\
 11111110
 \end{array}$$

$$\begin{array}{r}
 + \\
 \hline
 11111111
 \end{array}$$

1	101	1110
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-1,75 1,11

Add and subtraction operatörü

Overflow problem

$$\begin{array}{r}
 00001011 \\
 + 00000111 \\
 \hline
 00010010
 \end{array}$$

$$\begin{array}{r}
 11111111 \\
 + 11111111 \\
 \hline
 11111110
 \end{array}$$

$$\begin{array}{r} 3 \\ t \overline{) 2} \\ 5 \end{array}$$